P vs NP as Conjecture-Grade UQFF Anchor: Informational Buoyancy Argument and Audit Closure

Daniel T. Murphy

2026-05-17

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Abstract

We close audit gap #11 of the UQFF calibration program by explicitly registering the P vs NP problem as a CONJECTURE-tier entry with zero observational anchors. The UQFF informational-buoyancy stance asserts that exhaustive search has positive vacuum (SCm) buoyancy cost, $C_{\text{search}}(n) = B_{\text{inf}} \cdot 2^n$, which diverges from any polynomial alternative $C_{\text{solve}}(n) = B_{\text{inf}} \cdot n^k$ at $n \to \infty$, formally suggesting P $\neq$ NP. We make this conjecture explicit and recordable in the calculator registry, but emphasize that no falsifiable observational anchor exists. Calculator registered as `cp4_id = 445`; 20/20 smoke tests pass.

1. Motivation

Audit entry #11 noted that the UQFF framework had no formal handling of conjecture-grade mathematical claims with no observational anchor, breaking the G6 SM Anchor Gate audit trail. The fix is structural: explicitly mark such entries as CONJECTURE with `anchor = None`.

2. Informational Buoyancy Cost Functions

For an NP-complete decision problem of size n:

$$C_{\text{search}}(n) \;=\; B_{\text{inf}} \cdot 2^n, \quad C_{\text{solve}}(n) \;=\; B_{\text{inf}} \cdot n^k, \quad k \in \mathbb{N}.$$

UQFF buoyancy non-negativity requires

$$\boxed{\Delta_C(n) \;\equiv\; C_{\text{search}}(n) - C_{\text{solve}}(n) \;=\; B_{\text{inf}}(2^n - n^k) \;\geq\; 0 \;\forall n \geq n_*.}$$

By contrapositive (informal): if P = NP existed then $\Delta_C \to 0$ for all n, in tension with the asserted positivity of C_{search}. Hence the UQFF conjectural stance is P $\neq$ NP.

This argument is conjectural and has no observational anchor.

3. Statement Registry

Four formal statements are exposed in `STATEMENTS`:

ID	Name	Grade	Anchor	UQFF Position
S1	$P \neq NP$	conjecture	None	$\Delta_C > 0\ \forall n \Rightarrow P \neq NP$
S2	$NP \subseteq AM$	conjecture	None	orthogonal (no claim)
S3	$NP \neq$ co-NP	conjecture	None	proof/disproof informational asymmetry
S4	NP-intermediate exists (Ladner 1975)	conjecture	None	consistent, not asserted

4. UQFF Aether Modulation

Even though the P vs NP problem is purely mathematical, the calculator returns a nominal aether factor $f_A = 1 + \beta_i(\rho_{\mathrm{SCm}}/\rho_{\mathrm{amb}})\cos(\pi t_n)$ for registry-consistency; it does not affect the conjectural verdict.

5. Anchor Validation (Tier = CONJECTURE)

Anchor	Type	Verdict
—	none	n_anchors = 0; n_statements = 4; all unanchored

This is the correct outcome for a conjecture-tier entry. The calculator's `query_result` exposes `tier = "CONJECTURE"` so downstream audit tools can filter conjectural entries from observational anchor counts.

6. Code Registration

Module `_session301_pnp_conjecture_anchor.py`; class `PvsNPConjectureAnchor`; `cp4_id = 445`. Registered in `CondensedPhysics3.py` under `SESSION_301_CALCULATORS`. Smoke-test result: 20 / 20.

7. Falsifiable Predictions

(i) Any future proof of $P = NP$ would falsify the UQFF informational-buoyancy stance.
(ii) The conjecture-tier flag must propagate unchanged in all downstream UQFF audit reports.

8. Status

[CLOSED] audit gap #11 cp4_id = 445 session = 301 tier = CONJECTURE.

9. References

Cook 1971; Karp 1972; Ladner 1975. Clay Mathematics Institute Millennium Prize Problems. UQFF_CALIBRATION_AUDIT.md.